Problem 1. Consider the following function whose input is an undirected graph G. Edges of G are represented by an adjacency LIST. Assume that:

1. `Q.Insert`(x) takes $\Theta(\log(s))$ time,
2. `Q.ExtractMin`(x) takes $\Theta(\log(s))$ time.

where s is the number of elements in the data structure.

```
function Func(G)
    Q.Init()
    foreach vertex v_i ∈ G.V do
        foreach edge (v_i, v_j) ∈ G.E incident on v_i do
            Q.Insert(w(v_i, v_j))
        end for
    end for
    foreach edge (v_j, v_k) ∈ G.E do
        x = Q.ExtractMin()
        Print x
    end for
end function
```

$s = 1$

$s = 2m$

$s = 2m - m = m$

# of inserts:

$$\sum_{i=1}^{n} \sum_{j=1}^{\deg(v_i)} 1 = \sum_{i=1}^{n} \deg(v_i) = 2m$$

# of extracts:

$$\sum_{i=1}^{m} 1 = m$$

Let n be the number of vertices and m be the number of edges of G. Analyze the run time of the function and determine its complexity in terms of n and m.

$$T(n,m) = \sum_{s=1}^{2m} c\log(s) + \sum_{s=m}^{2m} c\log(s)$$

$$\left.\begin{aligned} &\le \sum_{s=1}^{2m} c\log(2m) + \sum_{s=m}^{2m} c\log(2m) \\ &\le (2m)\, c\log(2m) + (2m - m + 1)\, c\log(2m) \end{aligned}\right\} O(m\log(m))$$

$$\left.\begin{aligned} &\ge \sum_{s=m}^{2m} c\log(m) \\ &\ge (m+1)\, c\log(m) \end{aligned}\right\} \Omega(m\log(m))$$

Thus, $T(n,m) \in \Theta(m\log(m))$.

Problem 2. Consider the following function whose input is an undirected graph G. Edges of G are represented by an adjacency LIST. Assume that:

1. `Q.Insert`(x) takes $\Theta(\sqrt{s})$ time,
2. `Q.ExtractMin`(x) takes $\Theta(s)$ time.

where s is the number of elements in the data structure.

```
function Func(G)
    Q.Init()                                   s = 1
    foreach vertex v_i ∈ G.V do
        foreach vertex v_j ∈ G.V do
            Q.Insert(w(v_i, v_j))
        end for
    end for                                    s = n^2
    foreach edge (v_j, v_k) ∈ G.E do
        x = Q.ExtractMin()
        Print x
    end for                                    s = n^2 - m
end function
```

Let n be the number of vertices and m be the number of edges of G. Analyze the run time of the function and determine its complexity in terms of n and m. Assume $m \geq n$

$$T(n,m) = \sum_{s=1}^{n^2} \sqrt{s} + \sum_{s=n^2-m}^{n^2} s$$

$$\leq \sum_{s=1}^{n^2} \sqrt{n^2} + \sum_{s=n^2-m}^{n^2} n^2$$

$$\leq n^2 \cdot n + (m+1)n^2$$

$$\geq \sum_{s=\frac{n^2}{2}}^{n^2} \sqrt{n^2/2} + \sum_{s=n^2-m}^{n^2} (n^2-m)$$

$$\geq \frac{n^2}{2} \cdot \frac{n}{\sqrt{2}} + (m+1)(n^2-m)$$

$$\geq \frac{n^3}{2\sqrt{2}} + m \cdot \left(n^2 - \frac{n(n-1)}{2}\right)$$

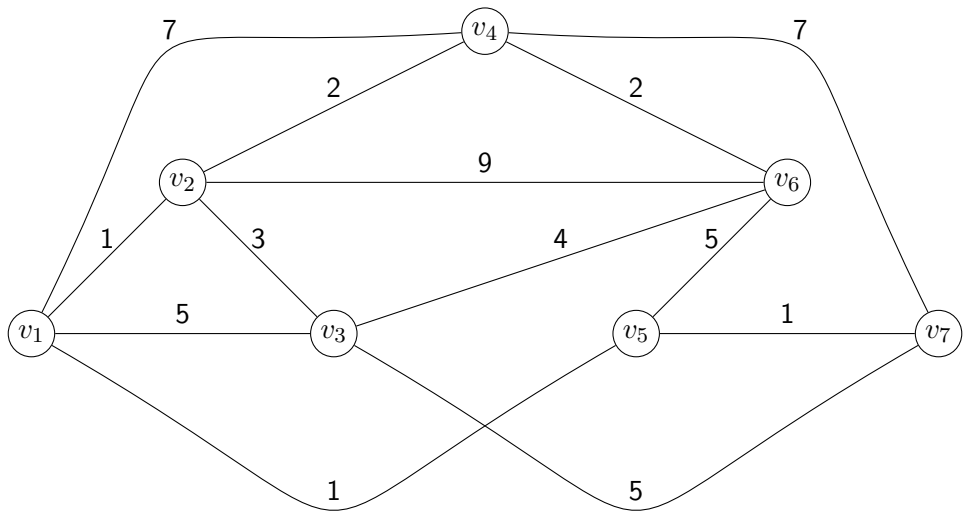
Thus, $T(n,m) \in \Theta(n^3 + n^2 m)$

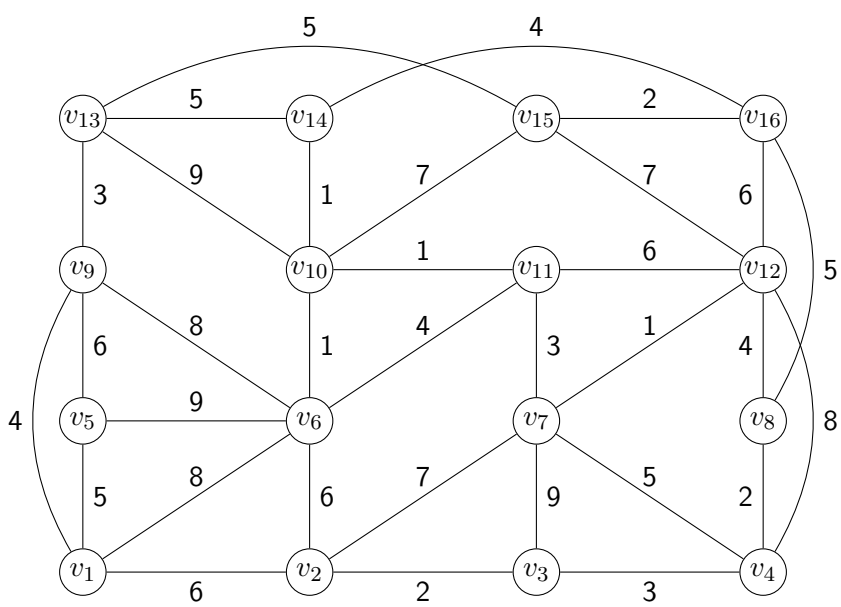
Problem 3. For each of the graphs given below,

(a) Find a minimum spanning tree of the graph.

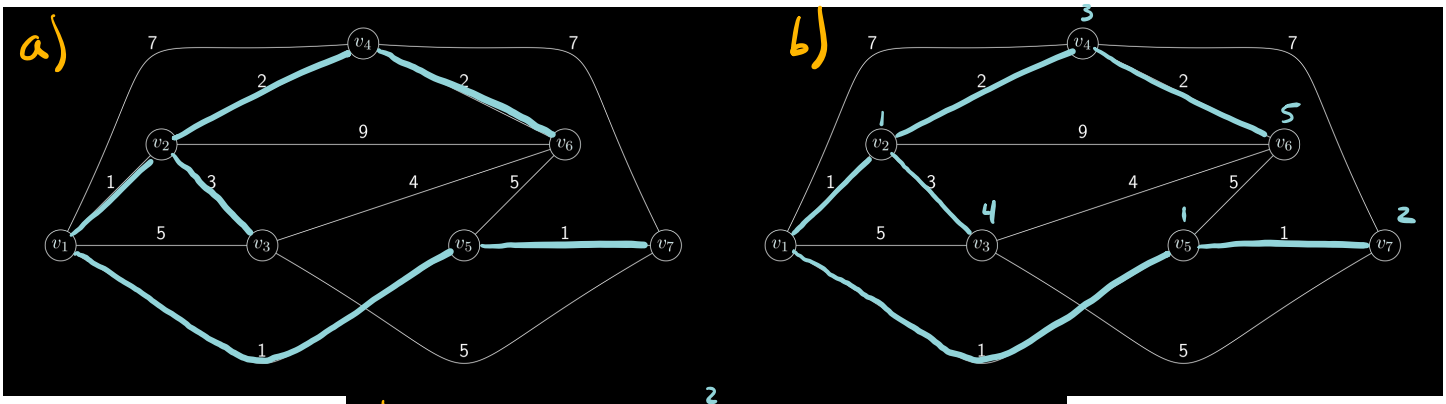
(b) Find a shortest path tree of the graph starting at vertex v_1.

(c) Find a shortest path tree of the graph starting at vertex v_6.





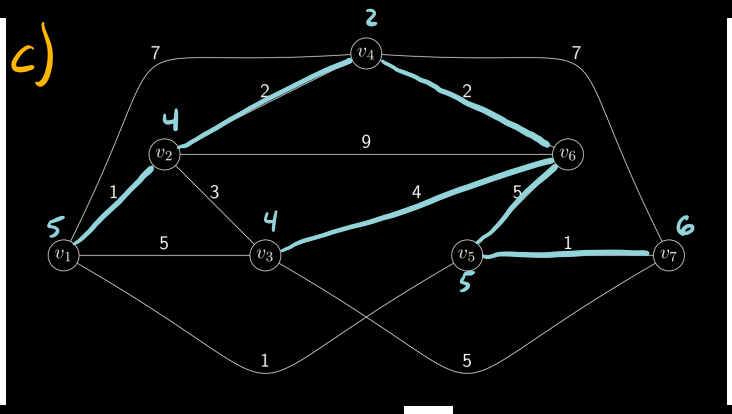
a)


v_1 v_2 v_3 v_4 v_5 v_6 v_7
7 7 2 2 9 1 3 4 5 5 1 1 5

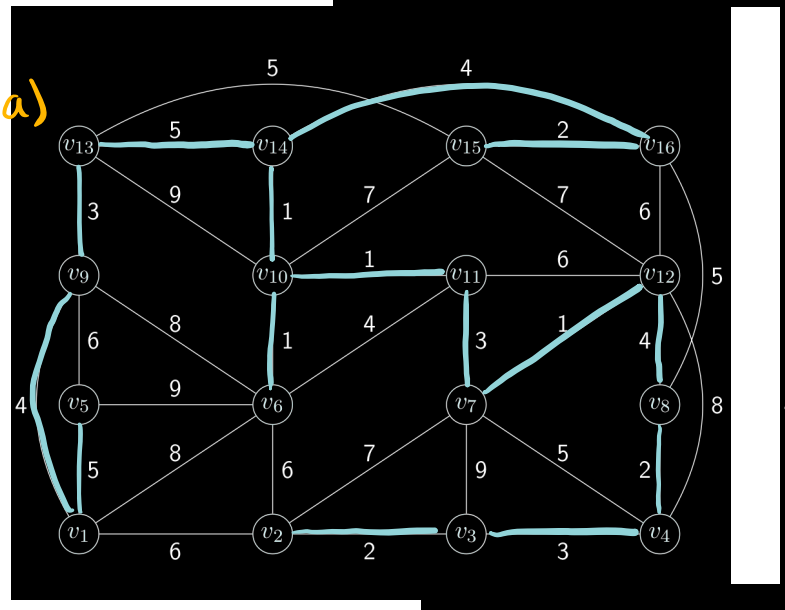
b)

v_1 v_2 v_3 v_4 v_5 v_6 v_7
3 1 5 4 1 2
7 7 2 2 9 1 3 4 5 5 1 1 5

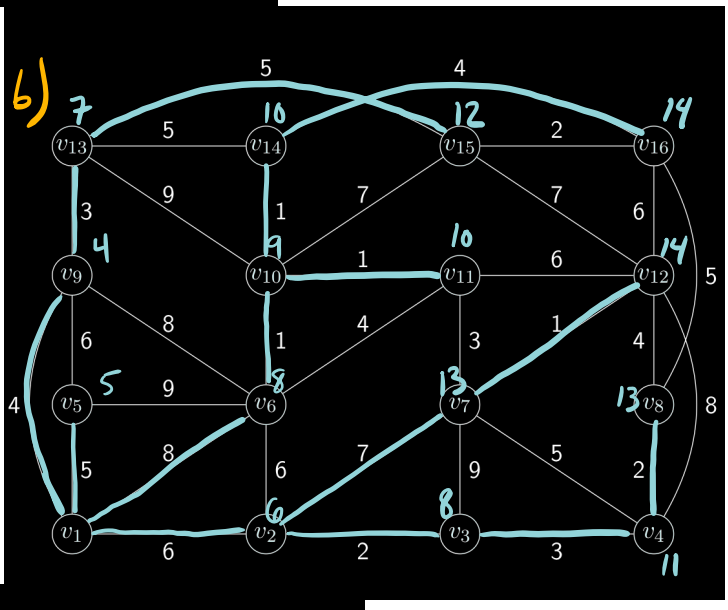
c)


v_1 v_2 v_3 v_4 v_5 v_6 v_7
2 4 5 4 5 6
7 7 2 2 9 1 3 4 5 5 1 1 5

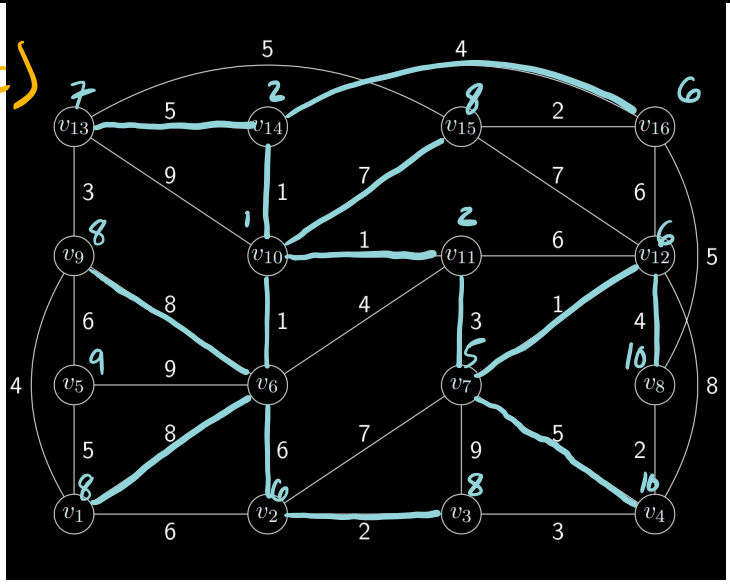
a)


v_1 v_2 v_3 v_4 v_5 v_6 v_7 v_8 v_9 v_{10} v_{11} v_{12} v_{13} v_{14} v_{15} v_{16}
5 4 5 2 3 9 1 7 7 6 1 6 5 6 8 1 4 3 1 4 4 9 8 5 8 6 7 9 5 2 6 2 3

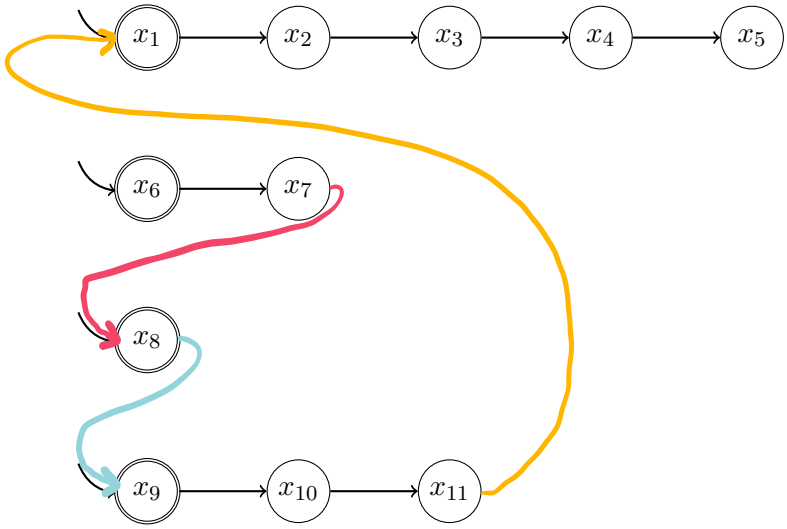
b)


v_1 v_2 v_3 v_4 v_5 v_6 v_7 v_8 v_9 v_{10} v_{11} v_{12} v_{13} v_{14} v_{15} v_{16}
7 10 12 14 4 9 10 14 5 8 13 13 6 8 11
5 4 5 2 3 9 1 7 7 6 1 6 5 6 8 1 4 3 1 4 4 9 8 5 8 6 7 9 5 2 6 2 3

c)


v_1 v_2 v_3 v_4 v_5 v_6 v_7 v_8 v_9 v_{10} v_{11} v_{12} v_{13} v_{14} v_{15} v_{16}
7 2 8 6 8 1 2 6 9 5 8 10 8 6
5 4 5 2 3 9 1 7 7 6 1 6 5 6 8 1 4 3 1 4 4 9 8 5 8 6 7 9 5 2 6 2 3

Problem 4. The following linked lists represent sets,



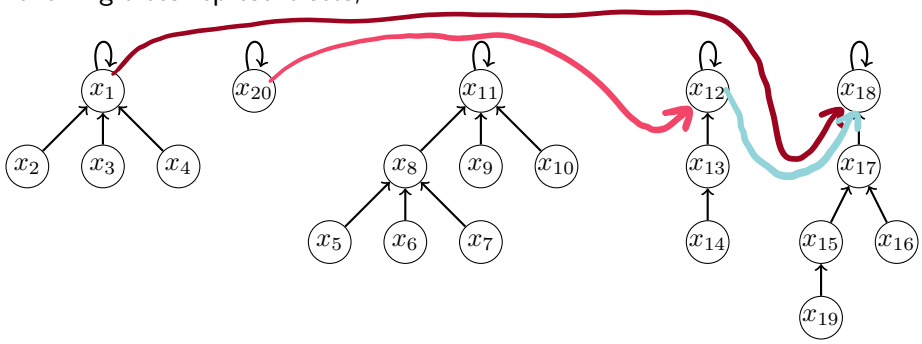
Consider the following sequence of Union-Find operations,

1: `WeightedUnion`(x_5,x_2) Nothing
2: `FindSet`(x_5) x_1
3: `FindSet`(x_3) x_1
4: `WeightedUnion`(x_6,x_8)
5: `FindSet`(x_6) x_6
6: `FindSet`(x_5) x_1
7: `WeightedUnion`(x_{11},x_8)
8: `FindSet`(x_{10}) x_6
9: `FindSet`(x_3) x_1
10: `WeightedUnion`(x_4,x_7)
11: `FindSet`(x_6) x_6
12: `FindSet`(x_2) x_6

1. What element is returned by each of the `FindSet` operations listed above?

2. Draw the linked lists representing the final data structure.

Problem 5. The following trees represent sets,



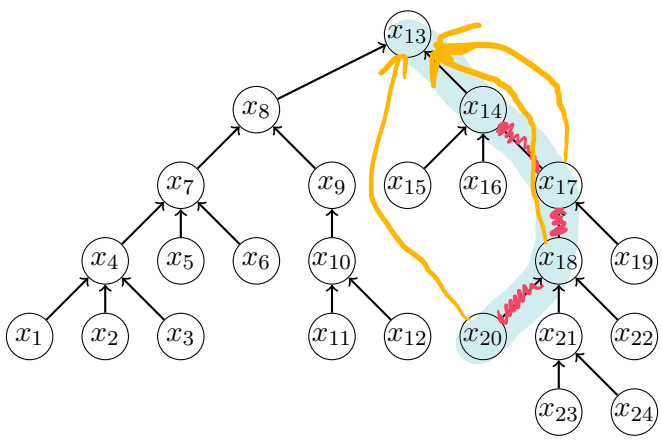
Consider the following sequence of Union-Find operations,

1: `UnionByHeight`(x_{18},x_4)
2: `FindSet`(x_4) X_4
3: `FindSet`(x_9) X_{11}
4: `UnionByHeight`(x_{20},x_{13})
5: `FindSet`(x_{14}) X_{12}
6: `UnionByHeight`(x_{18},x_{14})
7: `FindSet`(x_4) X_{18}

1. What element is returned by each of the `FindSet` operations listed above?

2. Draw the trees representing the final data structure.

Problem 6. Consider the following tree representing a set in the Union-Find data structure,



Run the `FindSet` operation on x_{20} with **path compression** and draw the resulting tree.

Problem 7. Consider the following Boolean expression

$$\varphi = (x_1 \vee x_2 \vee \overline{x}_4) \wedge (\overline{x}_1 \vee x_2 \vee \overline{x}_3) \wedge (x_1 \vee \overline{x}_3 \vee \overline{x}_4) \wedge (x_1 \vee \overline{x}_2 \vee x_4) \wedge (\overline{x}_1 \vee x_3 \vee \overline{x}_4) \quad (1)$$

1. Give the graph G corresponding to φ in the reduction of 3-SAT to the Independent Set problem.
2. Give an independent set of size 5 in G and a corresponding truth assignment which satisfies φ.

NOT ON EXAM!

Problem 8. Suppose that Q_1 can be solved in polynomial time and Q_1 can be reduced to Q_2 in polynomial time, can we say that Q_2 be solved in polynomial time?

$Q_1 \in P \rightarrow Q_1$ is easy

$Q_1 \leq_p Q_2$

NO!!

easy $\leq Q_2$

$A \leq_p$ easy $\rightarrow A$ is easy

P

hard $\leq_p B \rightarrow B$ is hard.

NP-C

SOLVE!!

Problem 9. Prove that each of the following problems are in P.

10-vertex cover Problem Given a graph G, does G have a vertex cover of size 10?

5-T Satisfiability Given a boolean formula φ with n variables, is there a truth assignment of $x_1, x_2, \ldots, x_n$ with exactly 5 variables true and the rest false such that φ is satisfied?

Vertex Distance Given an unweighted graph G and integer K, is there a pair of vertices u, v such that u and v are separated by at least K edges?

Periodic Array Given an array $A[]$ of size n, does there exist an integer T such that for $i = 1, 2, \ldots, T$,

$$A[i] = A[i+T] = A[i+2T] = \ldots A[i+n-T]?$$

For example,

$$A = [2, 7, 3, 4, 2, 7, 3, 4, 2, 7, 3, 4]$$

solves the problem with $n = 12$ and $T = 4$.

10-VC: foreach set S of 10 vertices
check if S is a vertex cover.

foreach edge:
check if edge touches a vertex in S

Runs $O(n^{10})$ times.
Body takes $O(m)$ time
$O(mn^{10})$.

5-T SAT: for each set S of 5 variables:
assign vars in S true
assign vars not in S false
$O(n)$
check if φ is true.
$O(m)$

Assume n vars. and m clauses.

Runs $O(n^5)$ times, so total $= O(n^6 + n^5 m)$

VD:

```
foreach vertex v:
    perform BFS starting at v.        } O(V+E)
    for each vertex u:                } O(V)
        if u.distance > K             } O(V+E)
            return true
        endif
    endfor
endfor
```

Takes $O(V^2 + VE)$ time

PA:

```
for T = 1, ..., N/2
    Check if A[i] = A[i+T] = ... = A[i+n-T]   } O(n)   } O(n^2)
```

Takes $O(n^2)$ time.

Problem 10. Prove that each of the following problems are in NP.

3-sum Problem Given an array of values A and integer K, are there values $A[i], A[j], A[k]$ in A that sum to K?

4-clique Problem Given a graph G, does G contain a clique of size 4?

Bipartite Graph Problem Given a graph G, is the graph bipartite?

Travelling Salesman Problem Given a weighted graph G, and an integer C, is there a cycle which visits every vertex with cost $\leq C$.

k-coloring Problem Given a graph G and integer k, can the graph be properly colored with $\leq k$ colors?

3Σ: Given A, K, x, y, z

1. Check if $x, y, z \in A$
2. Check if $x+y+z=K$.

$\Big\}\ O(n)$

4-clique: Given G, S

1. Check $|S| = 4$
2. foreach vertex $u \in S$
   foreach vertex $v \in S$
   check if $(u,v) \in E$

$\Big\}\ O(m)$

Bipartite: Given G, U, V:

1. for each edge $(x,y) \in G.E$
   check $x,y \in U$ <u>OR</u> $x,y \in V$

$\Big\}\ O(mn)$

TSP: Given G, C, P:

1. Check if Σ of weights of $P \leq C$. $\}\ O(n)$
2. Check P contains every vertex exactly once. $\}\ O(n^2)$

k-Coloring: Given G w/ colors.

1. Count # of colors $\leq K$ $\}\ O(n^2)$
2. for each edge $(u,v) \in G.E$
   check if $u.color = v.color$ $\Big\}\ O(m)$

Several of these can be done more efficiently

Problem 11. The Team Problem is:

> **Team Problem** Given a set of players S, a set of pairs of players who have practiced together, and an integer K, is it possible to form a team $S' \subseteq S$ of K players such that every pair of two members of the team S' have practiced together?

(a) Prove that the Team Problem is in NP.

(b) Prove that the Clique Problem can be reduced to the Team Problem in polynomial time.

(Note that proving both of these facts shows that the Team Problem is NP-Complete.)

An example of the Team Problem

$$S = \{\text{Alice, Bob, Charlie, Diya, Elle, Fred, Guanqi}\} \tag{2}$$

$$L = \{(\text{Alice,Diya}), (\text{Alice,Charlie}), (\text{Alice,Guanqi}), (\text{Charlie,Elle}), (\text{Diya,Guanqi}), (\text{Elle,Fred})\} \tag{3}$$

A valid team of 3 would be $S' = \{\text{Alice, Diya, Guanqi}\}$ because Alice has played with Diya and Guanqi, Diya has played with Alice and Guanqi, and Guanqi has played with Alice and Diya.

Let $S = V$

list $= E$

team $=$ clique

THESE PROBLEMS ARE NOT ON THE EXAM.
THESE PROBLEMS ARE NOT ON THE EXAM.

Problem 12. Consider the following two problems

Hamiltonian Cycle Problem Given a graph G, does G contain a cycle that visits every vertex exactly once?

Travelling Salesman Problem Given a weighted graph G, and an integer C, is there a cycle which visits every vertex with cost $\leq C$.

Prove that the Travelling Salesman Problem is NP-Complete by

1. Proving that the Travelling Salesman Problem is in NP.
2. Proving that the Hamiltonian Path can be reduced to the Travelling Salesman Problem in polynomial time.

   Hint: Given a graph, find a particular cost C and someway to assign weights to the graph to ensure that the unweighted graph has a Hamiltonian cycle if and only if the weighted graph has a cycle with weights $\leq C$.

Note: You will not be required to perform reductions like these, they are included for anyone who wants to see more of these problems before the end of the course.